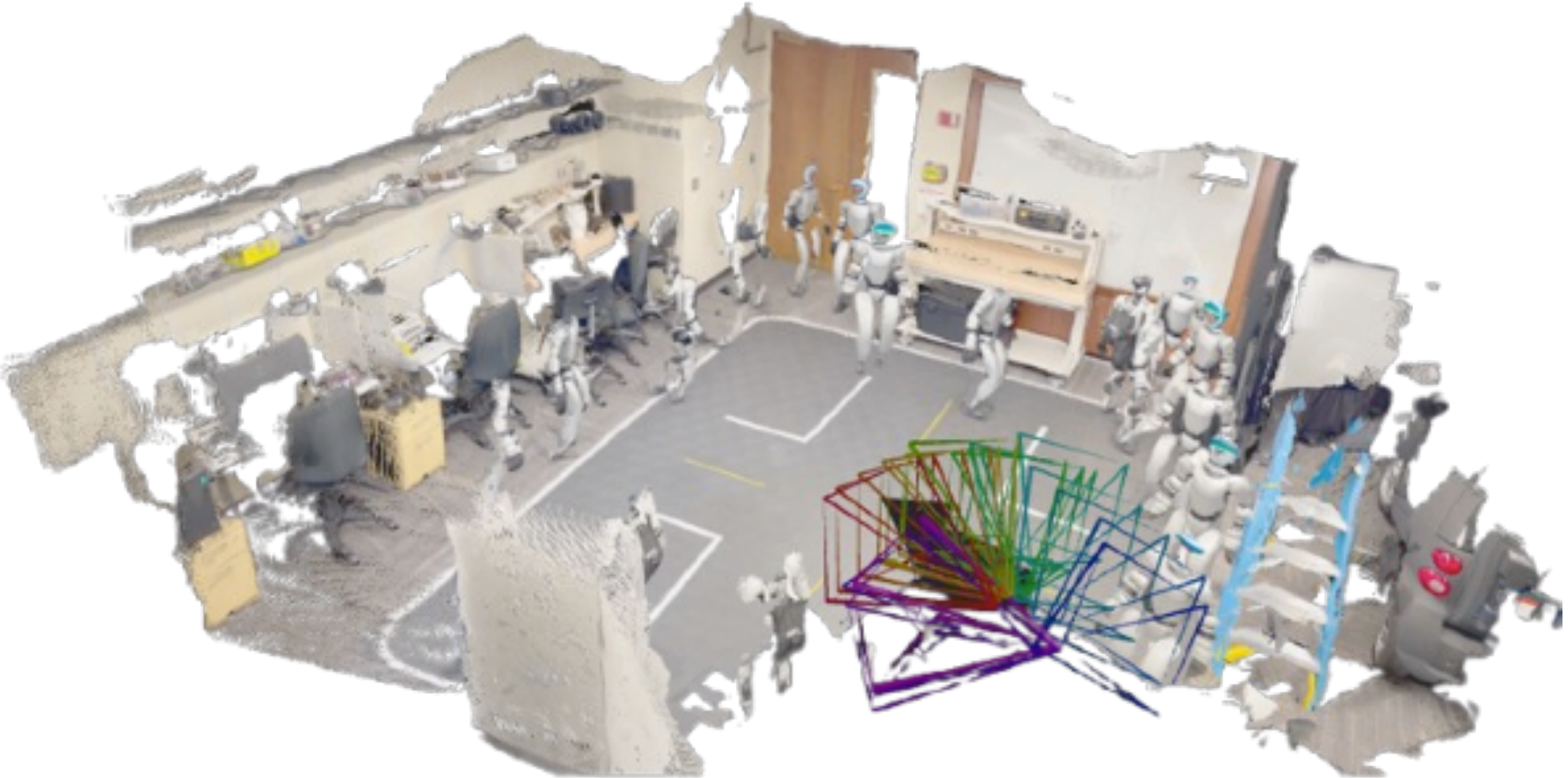
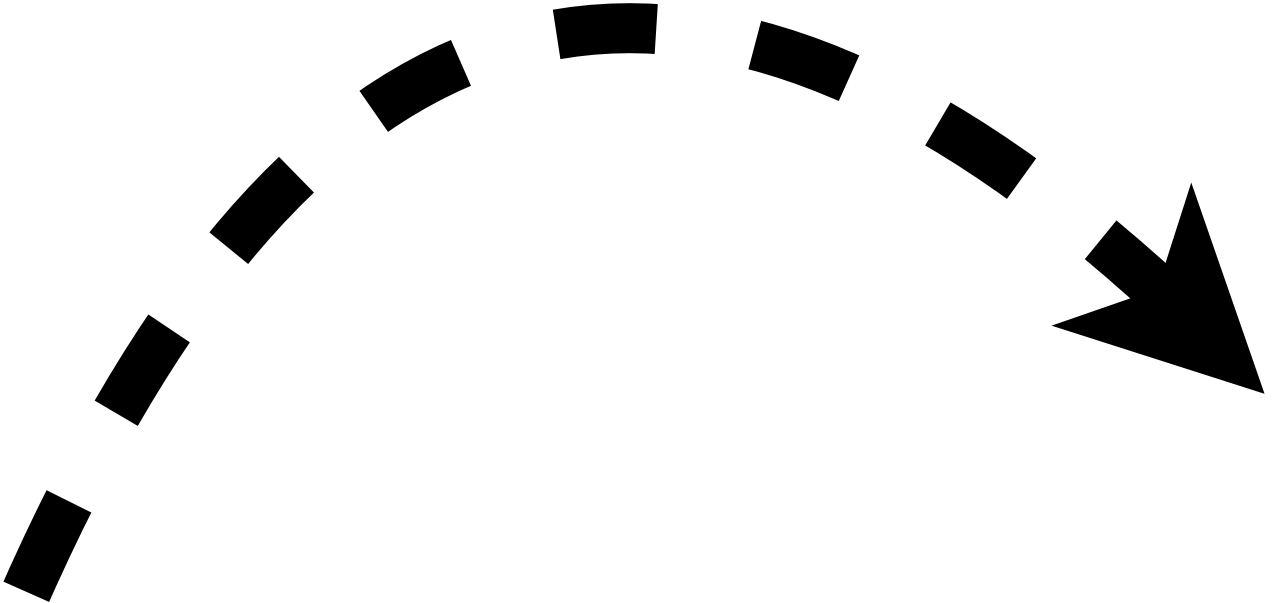
Spatial-Temporal Consistency & Long-term Memory

Plan 3: From Visual Intelligence to Physical AI



Long-term memory & planning



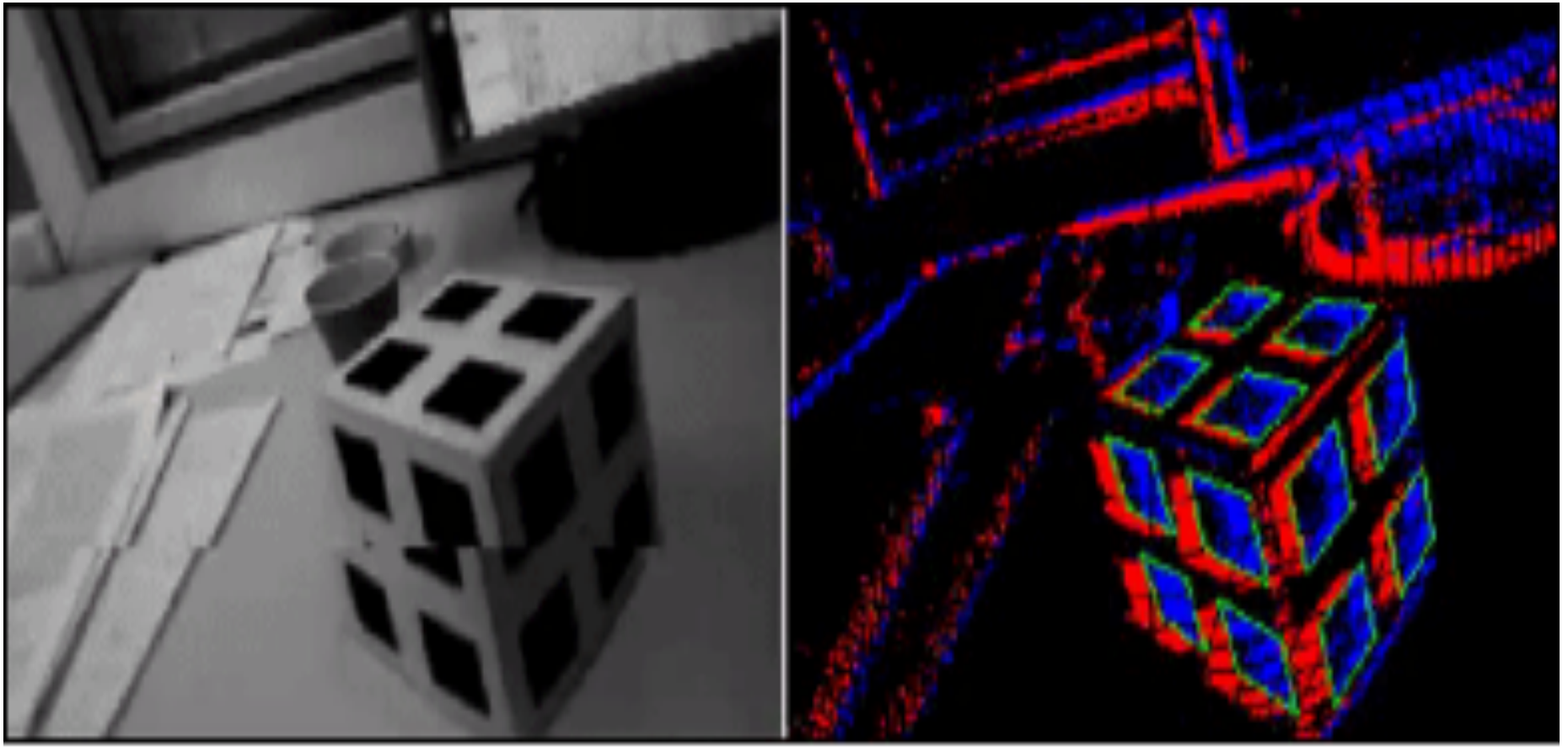
Action trajectory

Tom Williams

Spatial-temporal Consistency

Neil T. Bantam

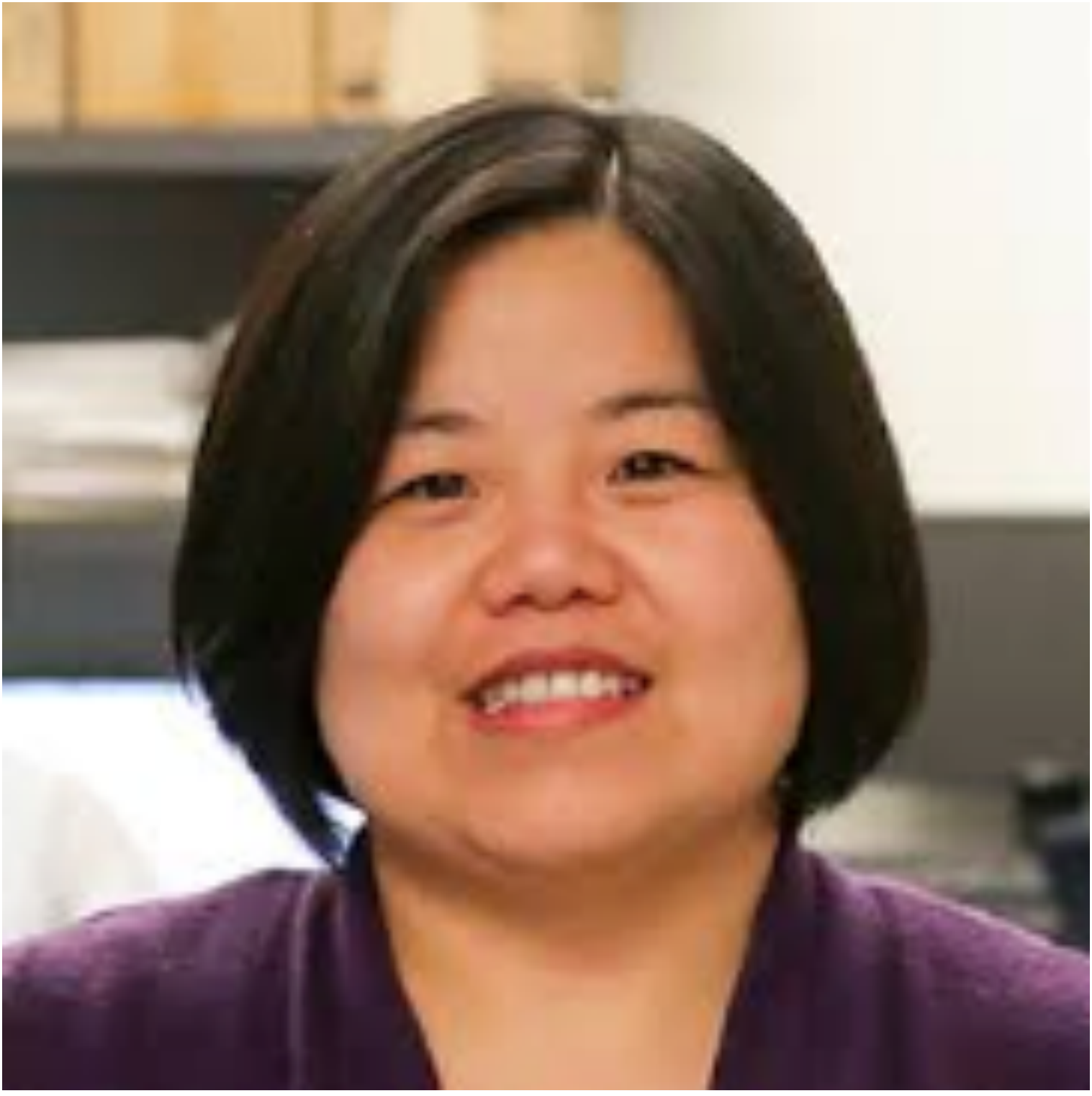
Runtime & Low Latency



Micah Corah

Kaveh Fathian

Xiaoli Zhang



@ Mines CS & ME & EE

NSF

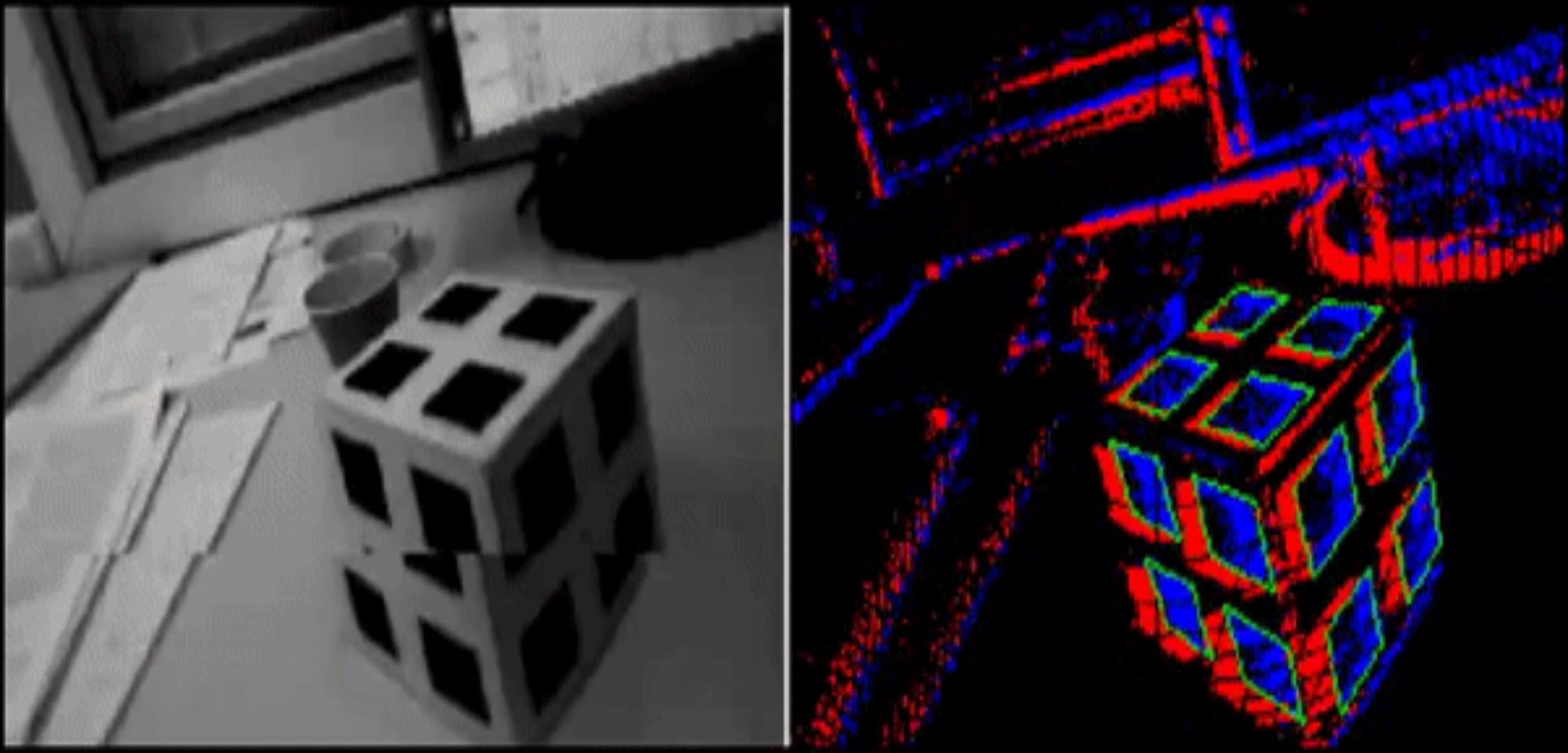
(Co-PI or PI)

Robust Intelligence (RI)

Foundational Research in Robotics (FRR)

CISE & ENG

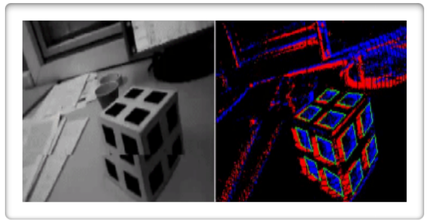
Estelle Smith



Plan 3: From Visual Intelligence to Physical AI

Spatial-Temporal Consistency & Long-term Memory

Runtime & Low Latency



Spatial-temporal Consistency

Long-term memory & planning

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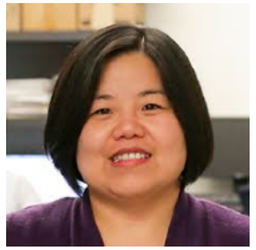
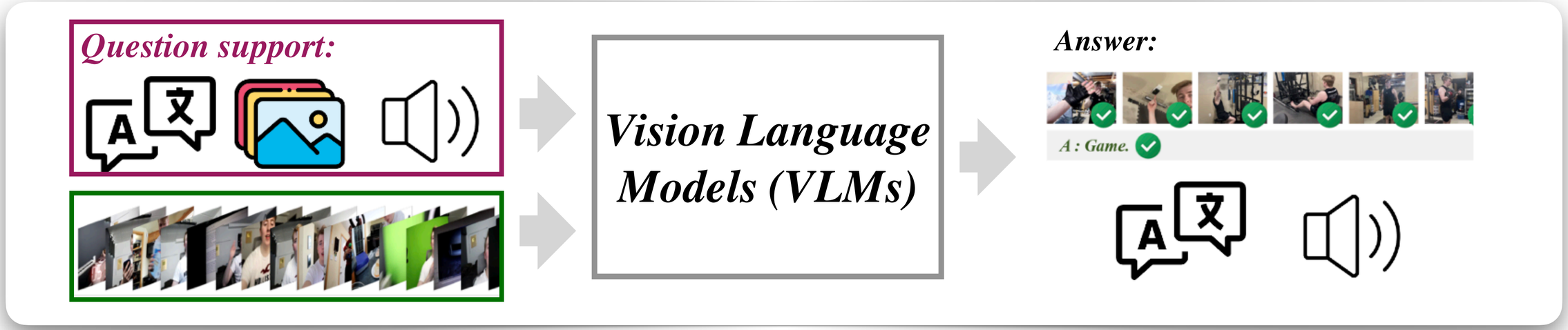


Xiaoli Zhang

@ Mines CS & ME & EE

Plan 4: Human-level Visual Intelligence



Key challenges

A. More Inteligent interaction

- *Language & visual & audio*
- *Motion-sensitive understanding*
- *Long-term causal reasoning*

B. Longer videos: 1-min, 15-min, 60-min, 2-hour, 2-day, 2-month

- *Limited Capacity (Model)*

NSF, CISE, Future CoRe

Robust Intelligence (RI)

- Small
- $500,000
- 3-year support

Faculty Early Career Development Program (CAREER)

- July, 2029-2031
- 5-year support